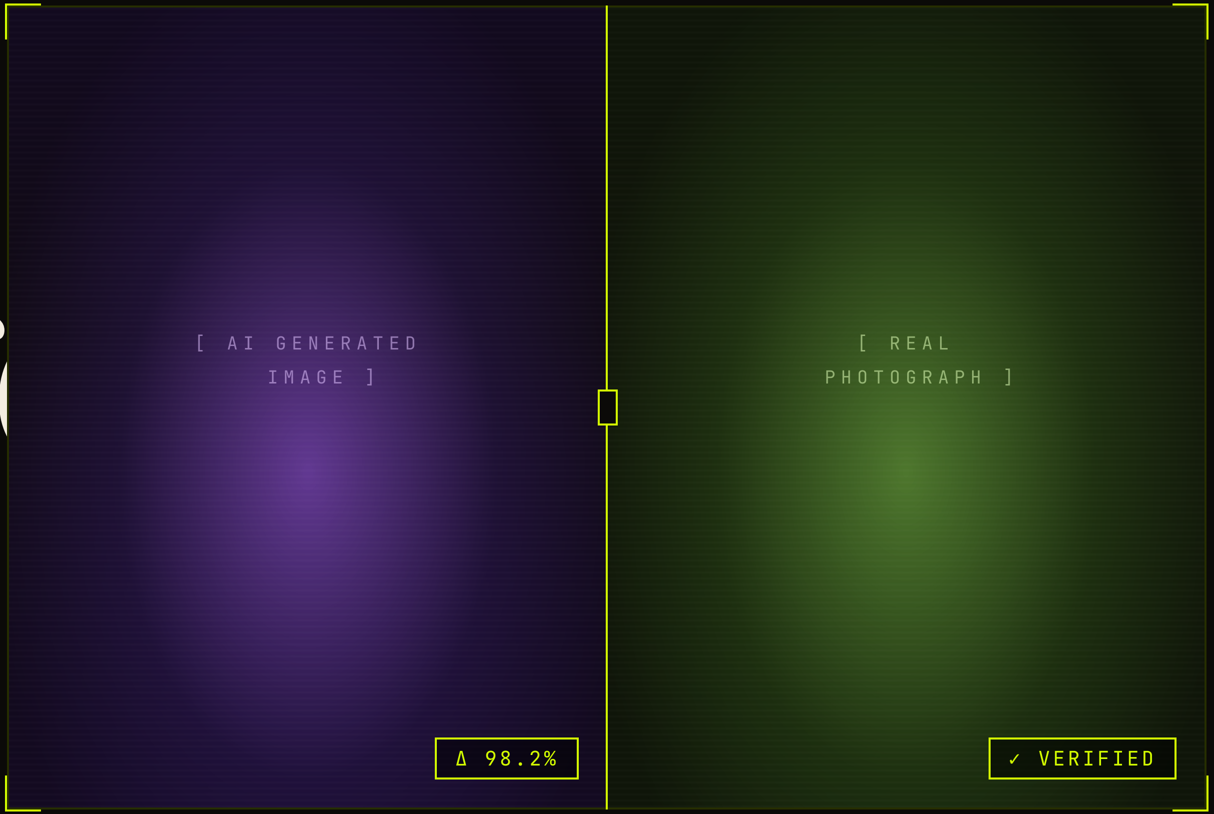
X 0412 · Y 0288 · Δ 98.2%

AUTHENTICITY REPORT

Is this image *real?*



● AI GENERATED ● REAL IMAGE

An MLOps pipeline for AI-generated vs. real image classification.

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ZHAW SCHOOL OF ENGINEERING — SPRING 2026

§ 01 GOAL & MOTIVATION

MLOps *pipeline* for AI-generated vs. real image classification.

CONTEXT

Generative models produce increasingly photorealistic images. Authenticity detection is a useful and well-scoped task that lets us focus on the **operations** layer of ML.

GOAL

Build a reproducible MLOps pipeline that classifies an input image as **real** or **ai-generated** with a confidence score.

IN SCOPE

Data versioning · experiment tracking · model registry · CI automation · containerized serving.

§ 02 DATA / SOURCE OF TRUTH

Every model is pinned to a *dataset version*.

SOURCE

AI Generated Images vs. Real Images

kaggle.com/datasets/tristanzhang32/ai-generated-images-vs-real-images

FAKE	30,000	REAL	30,000
Stable Diffusion	10,000	Pexels / Unsplash	22,500
MidJourney	10,000	WikiArt	7,500
DALL-E	10,000		

labels	real / fake
task	binary classification
splits	train / val / test
versioning	DVC
storage	external remote
license	academic use

kaggle

§ 03 PIPELINE / FLOW MAP

01 kaggle *Kaggle* ingest · raw/	02 *DVC* versioned · hashed	03 *PyTorch* train · EfficientNet-B0	04 *MLflow* metrics · registry
05 *GH Actions* CI · on push(main)	06 *ONNX* export · portable	07 *FastAPI + Docker* serve · /predict	08 *Streamlit* demo · UI

EACH MODEL ← (COMMIT · DATA-VERSION · RUN)

§ 04 COMPONENTS / ENTERED INTO EVIDENCE

DVC DATA VERSIONING Pins each snapshot to a commit.	*Git* SOURCE CONTROL Links runs to a commit SHA.	*PyTorch* TRAINING EfficientNet-B0 fine-tune.	*MLflow* TRACK & REGISTER Metrics, artifacts, stages.
GH Actions CI AUTOMATION Train → eval → register on push.	*ONNX* PORTABLE EXPORT Framework-agnostic inference.	*FastAPI* REST SERVING Lightweight endpoint for ONNX.	*Docker* CONTAINERIZATION Reproducible runtime.
Streamlit DEMO UI Interactive upload & prediction.	*GitHub* COLLABORATION Code, PRs, workflows.	*Kaggle* DATA SOURCE AI vs real image dataset.	[END OF LIST]

§ 05 WHY THIS IS MLOPS — NOT JUST ML

Any historical run can be *rebuilt.*

01 — **Reproducibility** — rebuild any past run from commit + DVC hash.

02 — **Traceability** — MLflow links metrics ↔ params ↔ artifacts ↔ data.

03 — **Automation** — CI retrains · evaluates · registers on every merge.

04 — **Deployability** — ONNX + Docker make the model portable.

05 — **Iterability** — new data → re-version → retrain → re-register.

EVERY MODEL = *(commit · data-version · run) — if any one is missing, the model is not reproducible.*